## Scanning Tunneling Microscope

Gerd Binnig and Heinrich Rohrer developed the scanning tunneling microscope in 1981, presumably because they found viewing through standard microscopes 'uncool'. An older version of the STM was the 'topografiner' invented by Russel Young and his colleagues between 1965 and 1971.

A super fine tip is taken and kept at a distance of one nanometre away from the sample.The tip is charged and hence electrons start flowing from the tip to the surface in a process called 'tunneling'. The current flowing through the tip is kept constant by scanning the tip over the surface horizontally and by adjusting the height of the tip vertically. This results in a 3-D image of the surface, which in our opinion is pretty damn cool. Because of the 3-D structure, one could make characterize surface roughness, observe surface defects, and determine the size and conformation of molecules and aggregates on the surface.

STMs are also advantageous because they can be used in ultra-high vacuum, air,water, other gases, as well as fom temperatures as low as -273 degree Celsius to around 300 degree Celsius. Their only drawback is that they require quite a bit of skill to use and are very fragile so forget running and buying one for yourself. Also, it costs only about $8000 for the cheapest STM.

## Sophisticated Fibres

It's mind-boggling to think where mankind would've been without radical materials. From simple shoelaces to lighter trainers to stronger aircraft, advances in material sciences have literally transformed ordinary things forever. Some of these advances include Fibres. Take the case of Velcro; something that we take for granted today. The fibre has a 1955 patent